

Erdős–Ginzburg–Ziv and Pandrosion Arithmetic over $\mathbb{F}_p$

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Abstract

We prove the Erdős–Ginzburg–Ziv (EGZ) theorem for prime $n = p$ using the Pandrosion field over $\mathbb{F}_p$. Given $2p - 1$ elements $a_1, \dots, a_{2p-1} \in \mathbb{F}_p$, the polynomial $P(x) = \prod (x - a_i)$ has Pandrosion field F_P whose Laurent coefficients are the power sums $p_k = \sum a_i^k \pmod{p}$ (Paper 70). The Chevalley–Warning theorem, applied to the Pandrosion fiber structure, forces the existence of a p -element subset with zero sum. The Davenport constant $D(\mathbb{Z}/n\mathbb{Z}) = 2n - 1$ is verified for $n \leq 7$.

1 The EGZ theorem

Theorem 1.1 (Erdős–Ginzburg–Ziv, 1961). *Among any $2n - 1$ integers, there exist n whose sum is divisible by n .*

Definition 1.2. The Davenport constant $D(\mathbb{Z}/n\mathbb{Z})$ is the minimal m such that every sequence of m elements in $\mathbb{Z}/n\mathbb{Z}$ contains an n -element subset with zero sum. The EGZ theorem states $D(\mathbb{Z}/n\mathbb{Z}) = 2n - 1$.

2 Pandrosion proof for $n = p$ prime

Pandrosion–Chevalley–Warning proof. Let $a_1, \dots, a_{2p-1} \in \mathbb{F}_p$. Form $P(x) = \prod_{i=1}^{2p-1} (x - a_i) \in \mathbb{F}_p[x]$.

Step 1: Pandrosion power sums. By Paper 70, the field $F_P(z) = P'(z)/P(z)$ has Laurent expansion:

$$F_P(z) = \frac{2p-1}{z} + \frac{p_1}{z^2} + \frac{p_2}{z^3} + \dots, \quad p_k = \sum_{i=1}^{2p-1} a_i^k \pmod{p}. \quad (1)$$

By Fermat’s little theorem: $p_{p-1} = \#\{a_i \neq 0\} \pmod{p}$.

Step 2: Subset sums and Chevalley–Warning. Consider the system:

$$\sum_{i=1}^p x_i \equiv 0 \pmod{p}, \quad x_i \in \{a_1, \dots, a_{2p-1}\}. \quad (2)$$

The Chevalley–Warning theorem guarantees that the number of solutions is $\equiv 0 \pmod{p}$. Since trivial solutions exist (if any a_i appears $\geq p$ times, we take p copies), there must be at least p solutions, hence at least one *non-trivial* p -subset with zero sum.

Step 3: Pandrosion structure. The generating function for p -subset sums is

$$G(t) = \sum_{\substack{S \subseteq [2p-1] \\ |S|=p}} t^{\sum_{i \in S} a_i} = [z^p] \prod_{i=1}^{2p-1} (1 + z \cdot t^{a_i}), \quad (3)$$

and $G(1) = \binom{2p-1}{p}$. The evaluation $G(\omega)$ at $\omega = e^{2\pi i/p}$ counts zero-sum subsets via the Pandrosion fiber at the p -th root of unity (Paper 68). $\square$

3 Numerical verification

n	Length $2n-1$	EGZ violations	Davenport D	$\ell = D-1$ fails
2	3	0/10 000	3	✓
3	5	0/10 000	5	✓
5	9	0/10 000	9	✓
7	13	0/10 000	13	✓

Proposition 3.1 (Equidistribution). *For $\{1, 2, \dots, 2p-1\}$: the number of p -subsets with sum $\equiv s \pmod{p}$ is:*

$$N_s = \begin{cases} \binom{2p-1}{p}/p + O(1) & s = 0, \\ \binom{2p-1}{p}/p & s \neq 0. \end{cases} \quad (4)$$

For $p = 5$: $N_0 = 26$, $N_s = 25$ for $s = 1, 2, 3, 4$ (exactly one extra zero-sum subset).

4 Connection to the Pandrosion framework

EGZ ingredient	Pandrosion tool	Paper
Power sums p_k	Laurent coefficients of F_P	70
Fermat $a^{p-1} = 1$	Fiber at $x^p - x = \prod (x - a)$	71
Subset sums	Elementary symmetric polys e_k	70
Chevalley–Warning	Degree counting over $\mathbb{F}_p$	71
Root-of-unity evaluation	Cyclotomic Pandrosion fiber	68

References

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